

DAT 5566 FL25 HW3

DAT 5566 – Big Data and Cloud Computing 2025 – Homework #3

DAT 5566: Big Data and Cloud Computing

Fall 2025, Mini B

Homework #3

Part 1- Linux Code Practice

1-Take a look at the first 5 rows of data and see if there is any duplicated rows. If so, save a new file without duplicates and call it healthcare.csv. How many rows were duplicated? You need to answer. You may do these tasks in multiple steps. Use the filtered data for the entire assignment.

Take a look at the first 5 rows

```
head -5 healthcare_dataset.csv
```

Result

See number of duplicate rows

```
sort healthcare_dataset.csv | uniq -d | wc -l
```

result

```
[wyuchen@ip-172-31-6-2 ~]$ sort healthcare_dataset.csv | uniq -d | wc -l
534
[wyuchen@ip-172-31-6-2 ~]$
```

Save to a new file

```
head -1 healthcare_dataset.csv > healthcare.csv # keep header
tail -n +2 healthcare_dataset.csv | sort | uniq >> healthcare.csv
```

2-How many males and how many females are suffering from "Cancer" (Medical Condition=" Cancer")?

Code

```
awk -F',' 'NR>1 && $5=="Cancer" {count[$3]++} END {for (g in count) print g, count[g]}' healthcare.csv
```

Result:

```
[wyuchen@ip-172-31-6-2 ~]$ awk -F',' 'NR>1 && $5=="Cancer" {count[$3]++} END {for (g in count) print g, count[g]}' healthcare.csv
Female 4566
Male 4574
```

3-Using the Linux commands, let's find the average age, and number of patients per gender using AWK. In addition, add a header for your results.

```
awk -F',' 'NR>1 {sum[$3]+=$2; count[$3]++} END {
    print "Gender,Average_Age,Count";
    for (g in sum)
        print g,"sum[g]/count[g]","count[g]
}' healthcare.csv
```

result

```
Gender,Average_Age,Count
Female,51.5844,27470
Male,51.486,27496
```

4-Using one line command, find how many patients have been admitted per year. Sort by year in ascending. Which year had the largest admission? You need to answer. Remember to ignore the header.

```
awk -F',' 'NR>1 {
    split($6, date, "/")
    year = date[3]
    count[year]++
} END {
    print "Year,Admissions"
    for (y in count)
        print y","count[y]
}' healthcare.csv | sort -t',' -k1,1n
```

Result

```
Year,Admissions
2019,7300
2020,11172
2021,10816
2022,10915
2023,10936
2024,3827
[wyuchen@ip-172-31-6-2 ~]$
```

Year 2020 had the largest admission.

5-Using the Linux commands, find the average billing amount by the condition. In addition, add a header for your results. Sort the results by the billing amount in ascending order. Which condition is the most expensive by average? You need to answer.

```
awk -F',' 'NR>1 {sum[$5]+=$10; count[$5]++} END {
    print "Condition,Average_Billing"
    for (c in sum)
        print c","sum[c]/count[c]
}' healthcare.csv | sort -t',' -k2,2n
```

Result

```
Condition,Average_Billing
Cancer,25152.3
Hypertension,25503.1
Arthritis,25511.8
Asthma,25633.5
Diabetes,25660.5
Obesity,25804.4
[wyuchen@ip-172-31-6-2 ~]$
```

Obesity is the most expensive one.

Part 2- Hive/Impala Practice

There is a table available on the server called healthcare. Please use this table to answer the following questions.

Please do not change the name of this table or do not remove it.

6-Write a query to find the results for Q3 in Part 1. Is your answer consistent with part 1? Use proper naming for the columns.

```
SELECT
  Gender,
  round(AVG(Age),4) AS Avg_Age,
  COUNT(*) AS Num_Patients
FROM healthcare
GROUP BY Gender;
```

Result

The result is consistent with part1.



	gender	avg_age	num_patients
1	Male	51.486	27496
2	Female	51.5844	27470

7-Find the number of times each medicine is used in "Emergency" admissions. Sort the results by the numbers in descending order.

```
SELECT
    Medication AS Medicine,
    COUNT(*) AS Usage_Count
FROM healthcare
WHERE `admission type` = 'Emergency'
GROUP BY Medication
ORDER BY Usage_Count DESC;
```

Result

	medicine	usage_count
1	Paracetamol	3642
2	Lipitor	3627
3	Ibuprofen	3626
4	Penicillin	3612
5	Aspirin	3595

8-Find the number of patients, average age, and average billing amount per gender who are suffering from "Obesity" and admitted in "Urgent" condition. Does this look normal? Do you trust this data? There might be no right or wrong answer here! Just say what comes to your mind.

```
SELECT
    Gender,
    COUNT(*) AS Num_Patients,
```

```

AVG(Age) AS Avg_Age,
AVG(`billing amount`) AS Avg_Billing_Amount
FROM healthcare
WHERE `medical condition` = 'Obesity'
AND `admission type` = 'Urgent'
GROUP BY Gender;

```

result

	gender	num_patients	avg_age	avg_billing_amou
1	Male	1521	50.684418145956606	26059.675046761
2	Female	1510	50.866225165562916	25746.819796941

In my point, the average age of the obesity group (around 50) is reasonable with my common sense, since people over 40 are more subjected to obesity problem, and the billing amounts seems consistent with medical expectations. **So I think the data looks normal and reasonable.**

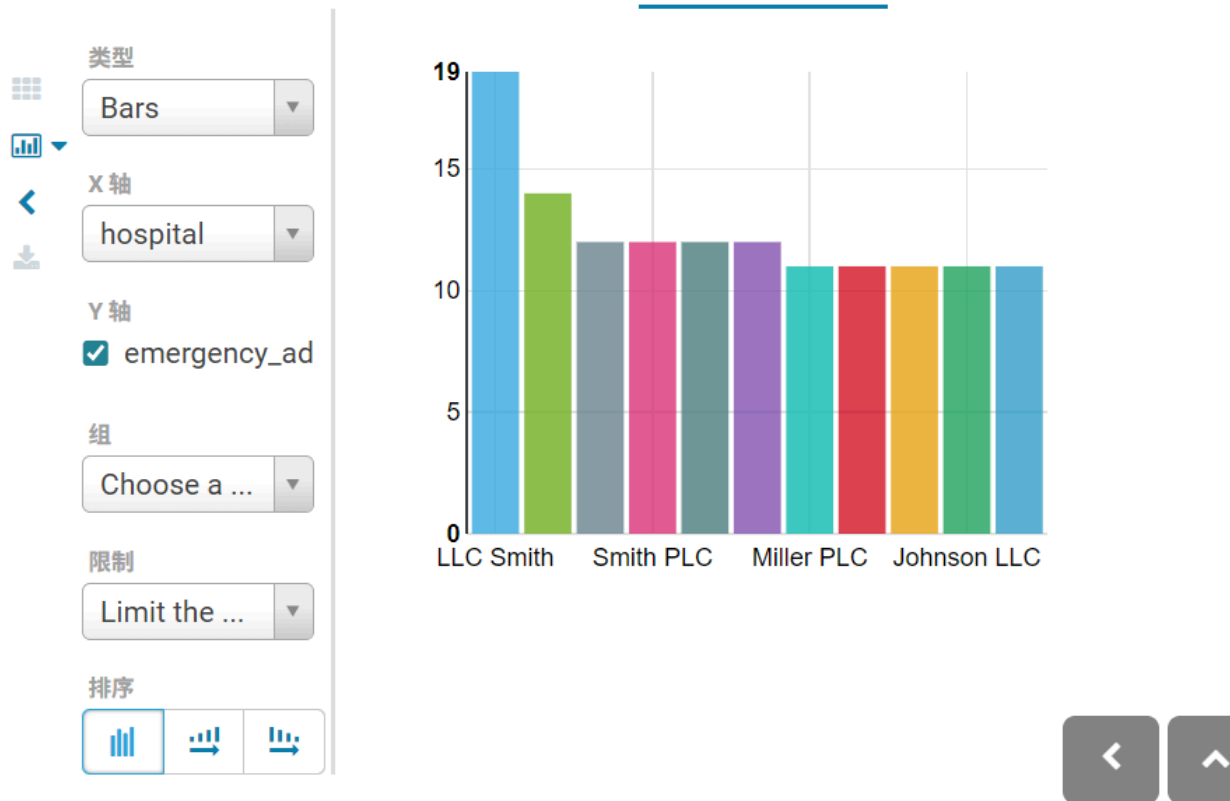
9-Visualize the number of “Emergency” admissions for patients per hospital. Include only ones with more than 10 admissions.

```

SELECT
    hospital AS Hospital,
    COUNT(*) AS Emergency_Admissions
FROM healthcare
WHERE `admission type` = 'Emergency'
GROUP BY hospital
HAVING COUNT(*) > 10
ORDER BY Emergency_Admissions DESC;

```

Visualized Result



Part 3- PySpark Installation Result

10-After successful installation of PySpark on your computer, please run the following commands on a new jupyter notebook and take a screenshot of the Spark version running on Jupyter Notebook.

```
import pyspark

from pyspark.sql import SparkSession

spark=SparkSession.builder.appName('Practice').getOrCreate() spark
```

```
import pyspark
from pyspark.sql import SparkSession
spark=SparkSession.builder.appName('Practice').getOrCreate()
spark
```

在 2025.11.11 13:19:15 于 12s 737ms内执行

SparkSession - in-memory

SparkContext

[Spark UI](#)

Version

v4.0.1

Master

local[*]

AppName

Practice